Mohammad Ashar Hussain

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SUMMARY

PhD candidate in Civil & Environmental Engineering at UW–Madison specializing in computational hydrology, stormwater infrastructure reliability, and machine learning-augmented simulation. Published researcher (4 papers) with expertise in Python, R, MATLAB, and SWMM — achieved 95% reduction in computational requirements using ML surrogate models and inverse FORM reliability analysis. Skilled in large-scale HPC workflows, satellite-derived hydrological data, and open-source scientific software development. Presented research at AGU 2023, 2024, 2025 and ASCE 2024.

EXPERIENCE

Graduate Research Assistant — University of Wisconsin–Madison | Madison, WI *Jan 2023 – Present*

- Refactored RainyDay, an open-source Python stochastic storm transposition (SST) tool → improved memory efficiency and reduced runtime via parallel programming, enabling scalable analysis of multi-terabyte MRMS radar-based precipitation datasets
- Applied inverse First Order Reliability Method (FORM) combined with ML surrogate models to replace computationally expensive Monte Carlo simulations → reduced computational requirements by 95% for large urban stormwater network reliability analysis
- Orchestrated large-scale SWMM hydrological simulations on CHTC high-performance computing cluster (HTCondor) → automated parameter sweeps across hundreds of storm scenarios with streamlined output aggregation pipelines
- Developed ML-based post-processing pipeline for crowdsourced urban rainfall categorizations → improved storm classification accuracy; published in Applied Computing and Geosciences
- Assessed design storm adequacy for large stormwater infrastructure using high-resolution MRMS radar precipitation data + SST → submitted manuscript to Journal of Hydrology proposing updated design validation standards

Research Assistant — Indian Institute of Technology Delhi | New Delhi, India *Jul 2022 – Dec 2022*

- Estimated river discharge using multi-source satellite imagery (NASA SWOT, Landsat, Sentinel) → developed remote sensing pipeline to derive streamflow without in-situ gauges, enabling hydrological monitoring in data-scarce regions
- Processed and analyzed satellite-derived surface water extent and hydraulic geometry relationships to calibrate discharge estimation models across multiple river cross-sections

EDUCATION

Ph.D., Civil & Environmental Engineering — UW–Madison | Madison, WI *Jan 2023 – Expected 2027*
Focus: Stormwater infrastructure reliability under extreme precipitation — radar-based SST & ML-augmented simulation | GPA: 3.6/4.0

M.Tech, Civil Engineering — Indian Institute of Science (IISc) | Bengaluru, India *2022*
GPA: 8.8/10 | Recipient: Half-Time Teaching/Research Assistantship (HTRA)

B.Tech, Civil Engineering — Jamia Millia Islamia | New Delhi, India *2019*
CGPA: 8.54/10

PUBLICATIONS

[Published] *Application of Machine Learning-Based Post-Processing to Improve Crowdsourced Urban Rainfall Categorizations* — Applied Computing and Geosciences (First Author)

[Under Review] *Assessing the Adequacy of Design Storms for Large Stormwater Networks using High-Resolution Weather Radar and Stochastic Storm Transposition* — Journal of Hydrology (First Author)

[Published] *Neglecting Spatiotemporal Rainfall Variability Underestimates Flood Hazard and Risk* — Natural Hazards (Co-Author)

[Under Review] *A Process-Informed Framework Linking Temperature-Rainfall Projections and Urban Flood Modeling* — Hydrology and Earth System Sciences (Co-Author)

CONFERENCE PRESENTATIONS

- American Geophysical Union (AGU) Fall Meeting — 2023, 2024, 2025
- American Society of Civil Engineers (ASCE) Annual Conference — 2024

TECHNICAL SKILLS

Languages & Tools: Python (NumPy, pandas, scikit-learn, multiprocessing/joblib), R, MATLAB, QGIS, Git, Bash
Hydrological Modeling: SWMM, RainyDay SST, MRMS radar precipitation processing, design storm analysis, HEC-RAS
Remote Sensing: NASA SWOT, Landsat, Sentinel-1/2, satellite-derived discharge estimation, GIS workflows
HPC & Computing: CHTC/HTCondor, parallel programming, large-scale batch simulation, workflow automation
Methods: Machine learning (classification, regression, surrogate modeling), inverse FORM reliability analysis, Monte Carlo simulation, stochastic hydrology, statistical analysis

HONORS & CERTIFICATIONS

GATE 2020 — Civil Engineering | Score: 778 | AIR: 632 (Top ~1% nationally) *2020*
Half-Time Teaching/Research Assistantship (HTRA) | Indian Institute of Science